

RAKESH KUMAR SWAIN

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EDUCATION

NIT Arunachal Pradesh

BTech in Computer Science & Engineering - 8.80 CGPA

2021 - 2025

WORK EXPERIENCE

ONGC

Full Stack Developer

June 2023 - July 2023

- Developed a Library Management System using **Flask** and **MySQL**, ensuring robust data storage and retrieval.
- Implemented key backend features such as **user authentication**, **book inventory management**, and **search functionality** to enhance user experience.
- Utilized **Git** for version control and collaborated with team members to perform **unit testing** and **debugging**, ensuring optimal performance and scalability.

PROJECT

Advanced Credit Card Fraud Detection Using Machine Learning Techniques

June 2023 - July 2023

Tool & Tech: Python, TensorFlow/Keras, Scikit-Learn, Pandas and NumPy, Matplotlib

- Engineered and evaluated **ANNs**, **XGBoost**, and **Random Forest** models for fraud detection with a focus on addressing dataset imbalance using **SMOTE** and other techniques.
- Achieved high model precision (**0.95**) and an F1-Score of **0.88**, while maintaining **99.96%** accuracy using XGBoost.
- Conducted data analysis to identify transaction patterns and provided actionable insights to improve fraud detection.

Chat With Multiple PDFs

Aug 2024 - Sep 2024

Tool & Tech: Python, Streamlit, Hugging Face, OpenAI API

- Developed a sophisticated chatbot using **LangChain**, facilitating **multi-PDF processing** and providing accurate, context-aware responses.
- Integrated **Hugging Face** and **OpenAI transformer models** for text-to-embedding conversion, enabling high-quality semantic similarity checks.
- Implemented **FAISS vector database** for efficient storage and rapid data retrieval.
- Deployed the solution using **Streamlit** and made it available as an open-source project on GitHub.

Text to Image Generator

March 2023

Tool & Tech: HTML, CSS, JavaScript, Hugging Face

- Built a web app leveraging **Hugging Face pre-trained models** to generate images from text input.
- Implemented real-time API calls for seamless user interaction and customized the application with responsive styles.

SKILLS

- **Programming Languages:** Python, SQL
- **Frameworks:** Flask (Basics)
- **Data Science and ML:** Data Analysis, Machine Learning, Deep Learning, NLP, Generative AI
- **Tools & Technologies:** Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, MS Excel
- **Other Skills:** Database Management System, OOP, Operating Systems, Computer Networking
- **Soft Skills:** Collaboration, Problem-solving, Communication

ACHIEVEMENTS

- Solved **400+ DSA problems** on LeetCode and GFG.
- Qualified for the **ML-Summer School 2024 program** by Amazon.